

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (CSE/ISE) (Credit System) Degree Examinations

Make up Examinations - May - June 2023

21CS303 / 21IS303 - COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain instruction execution and straight line sequencing with an example.	10	L*2	1	2
b) Define Clock rate and Basic performance Equation.	6	L2	1	2
c) For the following code compute program execution time. All the instructions have 4 clock cycles for the basic step and clock rate is 2 clock cycles per second. Assume that R1 contains 10, R2 indicates the first number and R0 is a temporary register to hold value.				
LOOP: ADD (R2), R0				
ADD #4, R2				
DECREMENT R1				
BRANCH>0 LOOP	4	L2	1	2
2. a) Explain the typical multiplication for the unsigned numbers with a neat diagram.	8	L2	2	2
b) Show how -13 X 11 is computed using sequential multiplier circuit.	6	L2	2	2
c) Make use of booth's recoding multiplier algorithm Multiply +13 and -6 (Multiplier).	6	L2	2	2
3. a) Explain the IEEE standard representation of floating point numbers with an example.	10	L2	2	1
b) Give the steps of non-restoring division and perform division where dividend is 1010 and divisor is 0011.	10	L2	2	2

Unit – II

4. a) Explain the single bus architecture with a neat diagram and its execution of an instruction using the architecture.	10	L2	3	1
b) Explain the multiple bus architecture with a neat diagram and its execution of an instruction in same architecture.	10	L2	3	1
5. a) Explain the control unit organization for the hardwired control with a neat diagram.	10	L2	3	2
b) Explain the block diagram of a complete processor.	10	L2	3	2
6. a) Explain the internal organization of the memory chip with a diagram.	10	L2	4	2
b) Distinguish between static and dynamic memory allocation in details.	4	L2	4	2
c) Write a short note for the following. i) Cache Hit ii) Cache Miss iii) Cache Coherence Problem	6	L2	4	2

Unit – III

7. a) Explain how input/output devices can be accessed with an example.	10	L2	5	2
b) Define an interrupt. Explain how to handle an interrupt with a diagram.	10	L2	5	2
8. a) Explain the concept of direct memory access (DMA) in a computer.	10	L2	5	2
b) Compare synchronous and asynchronous RAM cell.	5	L2	5	2
c) Write a short note for the following. i) Serial port ii) Parallel port	5	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
